

Notes: Pruitt & Wei (JF, 1989)

Institutional Ownership and Changes in the S&P 500

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1 Big picture

- **Question.** Are the abnormal returns around S&P 500 additions and deletions driven by actual institutional buying and selling pressure?
- **Context.** Shleifer (1986) and Harris and Gurel (1986) documented price and volume effects around changes in the S&P 500 and interpreted them as evidence that stock demand curves slope down.
- **Gap.** Earlier work inferred institutional demand from prices and trading volume. Pruitt and Wei directly examine changes in institutional ownership.
- **Data.** The paper uses Value Line institutional ownership data for 250 firms added to the S&P 500 from 1973 to 1986 and 17 non-bankruptcy/non-merger/non-tender-offer deletions.
- **Main result for additions.** Institutional ownership rises after S&P 500 additions. For the full sample, gross institutional ownership increases by 0.99 percentage points, and net ownership relative to control firms rises by 0.56 percentage points.
- **Time-series pattern.** The net increase is economically and statistically stronger in 1980–1986 than in 1973–1979, consistent with the growing importance of index funds.
- **Main result for deletions.** Institutional ownership falls after S&P 500 deletions. The net decline is 6.89 percentage points, with a t -statistic of -1.936 .
- **Return relation.** A cross-sectional regression shows that day-one abnormal returns after additions are positively related to net institutional ownership changes.
- **Economic takeaway.** The paper provides direct evidence that index changes affect institutional holdings. But the net institutional ownership changes are small relative to trading volume, suggesting that much of the liquidity demanded by index funds is supplied by other institutions.

2 Data and measurement

2.1 S&P 500 additions and deletions

- The addition sample contains 250 firms added to the S&P 500 from 1973 to 1986.
- The deletion sample contains only 17 firms because the authors exclude deletions caused by bankruptcy, mergers, and tender offers.
- The exclusion is important because these events can generate large ownership changes for reasons unrelated to index membership.
- The paper follows earlier S&P 500 price-pressure studies by comparing 1973–1979, 1980–1986, and the full 1973–1986 period.

Interpretation: The design is simple: test whether institutions actually increase holdings of additions and decrease holdings of deletions. The key novelty is direct institutional ownership data rather than price-only evidence.

2.2 Institutional ownership data

- Institutional holdings are obtained from *Value Line Investment Survey*.
- Value Line obtains holdings from SEC filings compiled by Viccars Data Services and Spectrum Disclosure Service.
- Because institutional filings are quarterly, the authors compare the calendar quarter before and the calendar quarter after addition or deletion.

- Firms with missing Value Line institutional ownership data are excluded.

Interpretation: Quarterly holdings are coarse. The paper cannot observe exact trading on the announcement day or effective day, but it can test whether index changes are followed by actual institutional rebalancing.

2.3 Institutional ownership measures

Let $I_{i,b}$ and $I_{i,a}$ be institutional holdings before and after the event, and let $T_{i,b}$ and $T_{i,a}$ be total shares outstanding before and after the event. The institutional ownership percentages are

$$\text{INST}_{i,b} = \frac{I_{i,b}}{T_{i,b}}, \quad \text{INST}_{i,a} = \frac{I_{i,a}}{T_{i,a}}. \quad (1)$$

The raw change is

$$\text{DIF}_i = \text{INST}_{i,a} - \text{INST}_{i,b}. \quad (2)$$

Interpretation: The raw change asks whether institutions own a larger or smaller fraction of the firm after the S&P 500 event.

2.4 Control adjustment

To control for economy-wide changes in institutional ownership, the authors subtract the mean change in institutional holdings for five Dow Jones 30 control firms:

$$\text{NDIF}_i = \text{DIF}_i - \text{MCDIF}_i. \quad (3)$$

The control firms are Eastman Kodak, Exxon, IBM, Minnesota Mining and Manufacturing, and Philip Morris.

Interpretation: The control adjustment matters because institutional ownership rose over time. Without adjustment, an apparent S&P 500 effect could partly reflect a broad secular shift toward institutional ownership.

3 Models and empirical specifications

3.1 Mean institutional ownership tests

For additions, the key null is

$$E[\text{DIF}_i] = 0 \quad \text{or} \quad E[\text{NDIF}_i] = 0.$$

For deletions, the price-pressure hypothesis predicts

$$E[\text{DIF}_i] < 0 \quad \text{and} \quad E[\text{NDIF}_i] < 0.$$

The authors report mean changes, standard errors, t -statistics, and the number of firms with institutional ownership increases.

Interpretation: Positive mean ownership changes after additions and negative changes after deletions are the direct ownership counterpart to earlier price-pressure evidence.

3.2 Binomial sign test

The paper also reports a proportionality test for the number of firms registering increases:

$$Z = \frac{P_o - 0.5}{\sqrt{0.25/N}}, \quad (4)$$

where P_o is the observed proportion of firms with positive institutional ownership changes and N is the number of firms in the sample.

Interpretation: The sign test asks whether ownership increases are more frequent than random coin flips. It is useful because ownership changes may be skewed.

3.3 Abnormal return regression

To connect holdings changes to price effects, the authors estimate a cross-sectional regression of the first post-listing abnormal return on net institutional ownership change:

$$AR_i = 0.88 + 3.43 \text{NDIF}_i, \quad (9.88) \quad (2.13), \quad (5)$$

where parentheses report t -statistics.

$$AR_i = \begin{matrix} 0.88 & + & 3.43 & (NDIF_i) \\ (9.88) & & (2.13). \end{matrix}$$

Figure 1: Original regression equation directly cropped from the paper.

Interpretation: The significant positive slope means that firms with larger institutional ownership increases also have larger abnormal returns. This links price pressure to actual institutional demand.

4 Identification and empirical design

4.1 What identifies the paper's result

- The event is addition to or deletion from the S&P 500.
- The outcome is the before-after change in institutional ownership.
- The control adjustment removes common movements in institutional ownership using five large Dow 30 stocks.
- The abnormal return regression connects post-listing price reactions to cross-sectional variation in ownership changes.

Interpretation: The design is not a modern causal design with random treatment. It is a direct event-based test of a mechanism proposed by earlier price-pressure papers.

4.2 Why the evidence supports price pressure

- Additions are followed by increases in institutional ownership.
- Deletions are followed by decreases in institutional ownership.
- The relationship between abnormal returns and net institutional ownership changes is positive.
- The strongest addition results occur in the later period, when index investing was becoming more important.

Interpretation: These four facts all point in the same direction: changes in index membership generate institutional demand shocks.

4.3 What the paper cannot fully prove

- It cannot separately identify index funds from other institutions in the Value Line holdings data.
- Holdings are observed quarterly, so the exact trading path around the event is not observed.
- The deletion sample is very small, with only 17 firms.
- The control group has only five firms, so it is a rough control for aggregate institutional ownership trends.
- The paper does not fully separate price pressure from all possible information effects around index changes.

Interpretation: The evidence is strongest as a direct mechanism check for earlier price-pressure studies, not as a complete decomposition of every motive for institutional trading.

5 Tables and figures

The paper contains no figures. The original tables below are directly cropped from the paper and grouped together to save space.

5.1 Table I and Table II: institutional ownership changes after additions and deletions

Read: Table I: additions

- For 1973–1979, gross institutional ownership rises by 0.45 percentage points, but net ownership relative to controls falls by 0.64 percentage points.
- For 1980–1986, gross institutional ownership rises by 1.45 percentage points, and net ownership rises by 1.61 percentage points.
- For the full 1973–1986 sample, gross ownership rises by 0.99 percentage points and net ownership rises by 0.56 percentage points.
- The full-sample net increase has a t -statistic of 2.08; the later-period net increase has a t -statistic of 4.08.
- The number of firms with ownership increases is also above chance in the full sample.

Read: Table II: deletions

- The deletion sample has only 17 observations.
- Gross institutional ownership falls by 4.85 percentage points.
- Net institutional ownership falls by 6.89 percentage points.
- The net deletion result has a t -statistic of -1.936 , which the authors interpret as suggestive at the ten percent level.

Economic message: The tables provide direct evidence that institutional investors rebalance after S&P 500 changes. Additions attract institutional ownership; deletions lose institutional ownership. The magnitude is not huge for additions, which implies high institutional demand elasticity and substantial liquidity supply by other institutions.

Table I
Mean Gross and Net Institutional Ownership Changes
Following Additions to the S&P 500 Index: 1973–1986

Panel A: Mean Institutional Ownership Changes Gross of Changes in the Control Firms (MDIF)^a

Time Period	Number of Firms	Mean Institutional Ownership Change (MDIF)	Standard Error	Test Stat. (t)	Number of Firms Registering Increase	Test Stat. (t)
1973–79	117	0.0045	0.0018	2.47	72	2.50
1980–86	133	0.0145	0.0039	3.76	78	1.99
1973–86	250	0.0099	0.0023	4.38	150	3.16

Panel B: Mean Institutional Ownership Changes Net of Changes in the Control Firms (MNDIF)^b

Time Period	Number of Firms	Mean Institutional Ownership Change (MNDIF)	Standard Error	Test Stat. (t)	Number of Firms Registering Increase	Test Stat. (t)
1973–79	117	−0.0064	0.0032	−2.00	59	0.09
1980–86	133	0.0161	0.0039	4.08	83	2.86
1973–86	250	0.0056	0.0027	2.08	142	2.15

^aThese tests compare the mean change in institutional holdings as reported in *Value Line* for the calendar quarters before and after addition to the S&P 500. (Positive values indicate increases in institutional holdings.)

^bThese tests compare the mean change in institutional holdings as reported in *Value Line*, net of changes in a control group of five Dow Jones 30 firms, for the calendar quarters before and after addition to the S&P 500. (Positive values indicate increases in net institutional holdings.)

(a) Table I: additions to the S&P 500.

Table II
Mean Gross and Net Institutional Ownership Changes
Following Deletions from the S&P 500 Index: 1973–1986

Panel A: Mean Institutional Ownership Changes Gross of Changes in the Control Firms (MDIF)^a

Time Period	Number of Firms	Mean Institutional Ownership Change (MDIF)	Standard Error	Test Stat. (t)	Number of Firms Registering Increase	Test Stat. (t)
1973–86	17	−0.0485	0.0341	−1.424	9	0.243

Panel B: Mean Institutional Ownership Changes Net of Changes in the Control Firms (MNDIF)^b

Time Period	Number of Firms	Mean Institutional Ownership Change (MNDIF)	Standard Error	Test Stat. (t)	Number of Firms Registering Increase	Test Stat. (t)
1973–86	17	−0.0689	0.0356	−1.936	9	0.243

^aThese tests compare the mean change in institutional holdings as reported in *Value Line* for the calendar quarters before and after deletion from the S&P 500. (Negative values indicate decreases in institutional holdings.)

^bThese tests compare the mean change in institutional holdings as reported in *Value Line*, net of changes in a control group of five Dow Jones 30 firms, for the calendar quarters before and after deletion from the S&P 500. (Negative values indicate decreases in net institutional holdings.)

(b) Table II: deletions from the S&P 500.

Figure 2: Original paper tables, grouped compactly.

6 Short summary

- Earlier studies showed price and volume effects around S&P 500 additions and deletions.
- Pruitt and Wei ask whether actual institutional ownership changes in the predicted direction.
- They use *Value Line* institutional holdings for 250 additions and 17 clean deletions from 1973 to 1986.
- Institutional ownership changes are measured before and after the event at the quarterly level.
- Net ownership changes subtract the average ownership change in five Dow Jones control firms.
- Additions are associated with positive institutional ownership changes, especially in 1980–1986.
- Deletions are associated with negative institutional ownership changes.
- First post-listing abnormal returns are positively related to net institutional ownership changes.
- The results support the price-pressure interpretation of index inclusion effects.
- The small size of the net ownership changes relative to trading volume suggests that non-index institutions help supply liquidity.

7 Conclusion

7.1 Core conclusion

Pruitt and Wei provide direct evidence that S&P 500 index changes are followed by institutional ownership changes in the expected direction. The paper strengthens the price-pressure interpretation of earlier abnormal return and volume studies by showing that institutional holdings actually move after additions and deletions.

7.2 Economic intuition

Index inclusion creates demand from investors who need to hold the S&P 500 portfolio. If demand curves for individual stocks are not perfectly elastic, that demand can temporarily or permanently affect prices. The

paper shows that institutions do increase ownership after additions and reduce ownership after deletions, and that these ownership changes are positively related to abnormal returns.

7.3 Takeaway

The paper is short but important because it turns the S&P 500 price-pressure argument from an indirect price-volume story into a direct institutional-ownership test. Its most interesting nuance is that the ownership changes are small relative to trading volume, implying that other institutions provide liquidity and that institutional demand for individual stocks may be highly elastic.